Chapter 2 Cells

Name:_____Class:_____Date: _____

EXTRA WORKSHEET 3: Plant systems

SCIENCE UNDERSTANDING **VISUAL/SPATIAL**

Cells, tissues and organs

Figures 1 and 2 below show images of parts of plants taken through a microscope. Photographs taken in this way are known as photomicrographs.

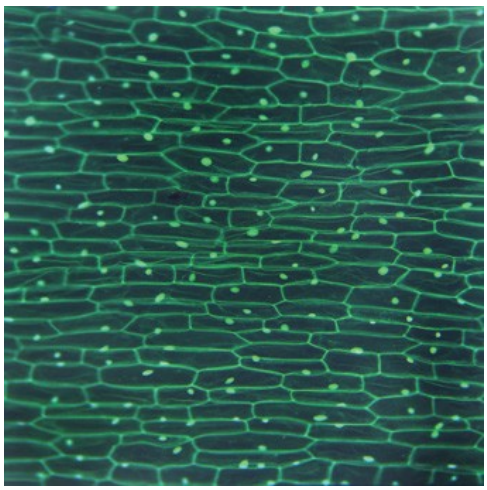


Figure 1

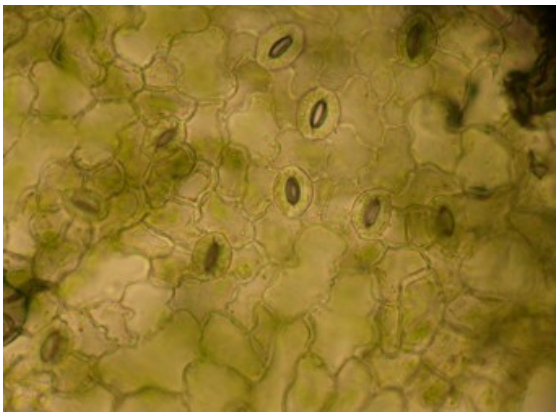


Figure 2

1 a **Deduce** whether Figure 1 is a photomicrograph of a cell, tissue or organ.

_____

b Justify your conclusion.

_____

_____

2 a **Deduce** whether Figure 2 is a photomicrograph of a cell, a tissue or an organ.

_____

b **Justify** your conclusion.

_____

_____

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Figure 3 below is a photomicrograph of a plant stem. The stem has been cut across as if cutting off a flower. If the complete stem could be seen it would appear as a circle.

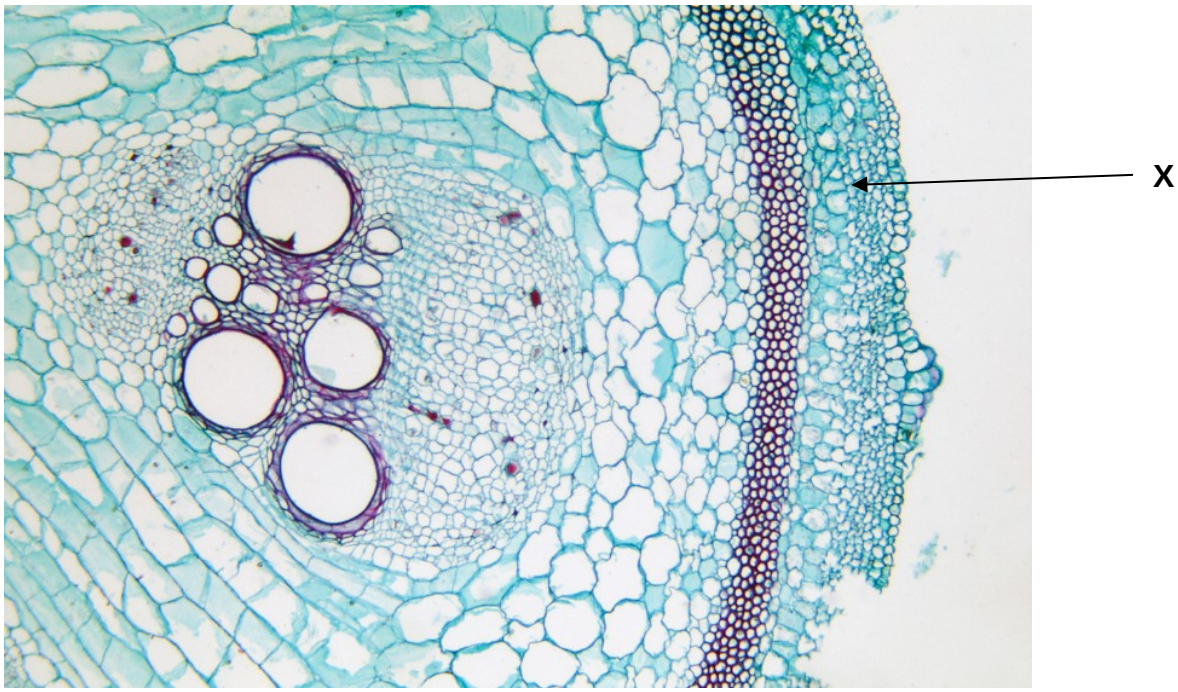


Figure 3

3 The stem is a plant organ. **Select** the information from the photomicrograph that supports this statement.

_____

_____

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4 **a** **Draw** an arrow on Figure 3 pointing to the part of the stem that is most likely to be the cells that carry water from the roots to the shoots.

b **Justify** why you placed the arrow in that position.

_____

_____

5 **a** **Propose** a function of the layer of cells marked with an X in Figure 3.

_____

b **Justify** your answer.

_____

Plant pathways

Figure 4 below is a leaf with the veins clearly showing.



Figure 4

6 **Draw** red arrows on Figure 4 to show the pathway of water through the leaf.

7 **Draw** blue arrows on Figure 4 to show the pathway of food through the leaf.

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